LAY/ERD

FINANCE FUNDAMENTALS · FORMULA GUIDE

The Math of Risk and Return

Every formula for measuring return and risk, for one investment and for a whole portfolio. Each comes with a plain-English meaning, a worked example, and the mistakes to avoid. All examples use made-up numbers.

A quick note: I'm not a finance professional. I'm sharing what I've learned, in plain language, so it's easier for the next person. Nothing here is financial advice.

How the pieces fit: measure what one investment earned (§1) → estimate what it might earn and how uncertain that is (§2, §3) → do the same for a basket of investments (§4, §5) → see how mixing investments changes risk (§6) → measure the risk each holding actually adds (§7).

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1. Return on one investment

What did this investment actually earn me over one period?

$$\text{Market value} = \text{units} \times \text{price per unit}$$

$$\text{Total profit} = \text{MV}_{\text{end}} - \text{MV}_{\text{start}} + \text{income received}$$

$$R = \frac{P_{\text{end}} - P_{\text{start}} + C}{P_{\text{start}}} = \frac{P_{\text{end}} - P_{\text{start}}}{P_{\text{start}}} + \frac{C}{P_{\text{start}}}$$

$$\text{Profit} = R \times \text{amount invested}$$

An investment pays you two ways: the **price moves** (capital gain or loss), and it may pay **income** along the way (a dividend, for a stock). Add both and divide by what you paid at the start.

Example. You buy 50 shares at $40 each ($2,000 invested). A year later they're worth $43 each, and each paid a $1.20 dividend.

Capital gain yield	($43 − $40) / $40 = **7.50%**
Dividend yield	$1.20 / $40 = **3.00%**
Total return	7.50% + 3.00% = **10.50%**
Profit	10.50% × $2,000 = **$210.00** (check: $2,150 − $2,000 + $60 = $210)

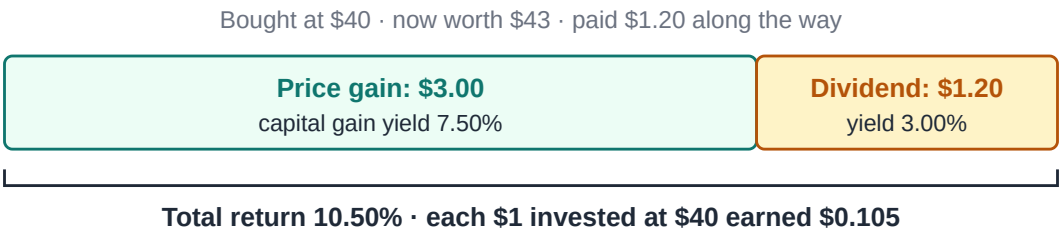


Figure 1. A return has two sources: the price moved, and cash came in.

When the share changes shape: spin-offs and splits

Sometimes one share turns into something else during the period: a company spins off a division as a new company, or splits its stock. The fix is the same both times: **follow one share you bought at the start**, and value everything it turned into.

$$R = \frac{\text{value now of everything 1 share became} - P_{\text{start}} + \text{dividends on it}}{P_{\text{start}}}$$

Spin-off	Bought 1 share at $50. It now trades at $46, and you also received 0.5 shares of the spun-off company, worth $10 each. Dividend $1. Value now = $46 + 0.5 × $10 = $51. R = ($51 − $50 + $1) / $50 = **4.00%**
2-for-1 split	Bought 1 share at $80. It became 2 shares at $42 each, and each paid $0.50. Value now = 2 × $42 = $84; dividends = $1. R = ($84 − $80 + $1) / $80 = **6.25%**

Watch out

- Divide by the **starting** price, never the ending price.
- After a split, don't compare the new $42 price to the old $80 price. That looks like a 47.5% loss but isn't one.
- Income counts. A stock that went nowhere but paid a dividend still earned a return.

2. Expected return and risk, when you know the possible outcomes

If I can list what might happen and how likely each outcome is, what should I expect, and how uncertain is it?

$$E\left(R\right) = \sum p_s R_s$$

$$\sigma^2 = \sum p_s \left(R_s - E\left(R \right) \right)^2 \quad \sigma = \sqrt{\sigma^2}$$

Expected return is the probability-weighted average of the outcomes: your best single guess. **Variance** takes each outcome's gap from that guess, squares it (so misses in either direction count), and weights it by its probability. **Standard deviation** (σ, "risk") is the square root of the variance, which puts it back in percent.

Example. A stock could have a strong year (+25%, 30% chance), a normal year (+10%, 50% chance), or a weak year (−15%, 20% chance).

Expected return	0.30 × 25% + 0.50 × 10% + 0.20 × (−15%) = 7.5% + 5% − 3% = **9.50%**
Variance	0.30 × (25 − 9.5)² + 0.50 × (10 − 9.5)² + 0.20 × (−15 − 9.5)² = 72.08 + 0.13 + 120.05 = **192.25 %²**
Risk (σ)	√192.25 = **13.87%**

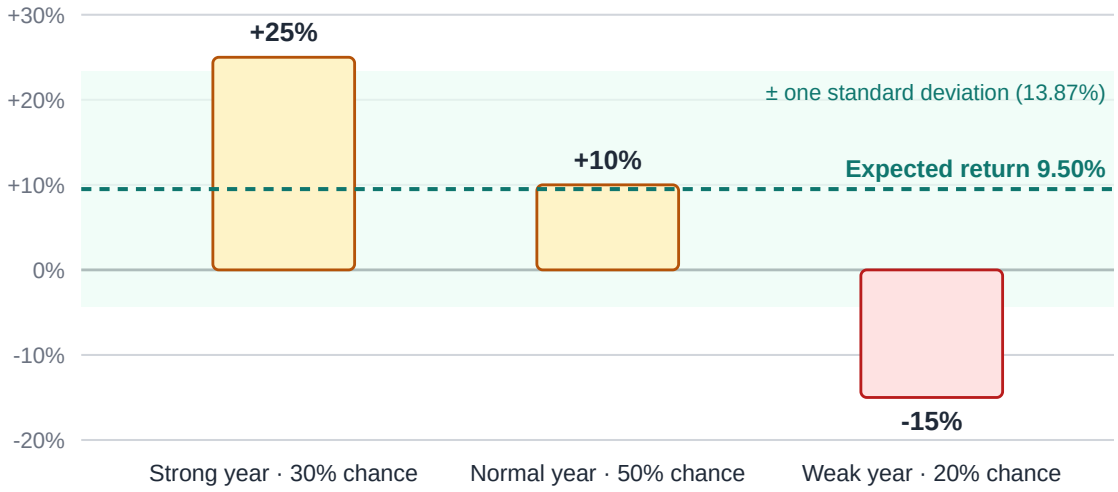


Figure 2. Three possible years. The expected return is the probability-weighted middle; risk measures how far outcomes typically land from it.

Watch out

- Probabilities must add up to 100%.
- Risk counts surprises in *both* directions. A big upside outcome raises risk too.
- Variance is in odd units (%²). Always take the square root before you interpret it.

3. Average return and risk from past returns

I don't know the probabilities, but I have several years of history. What's a typical year, and how bumpy is the ride?

$$\bar{R} = \frac{R_1 + R_2 + \cdots + R_T}{T}$$

$$\sigma^2 = \frac{\sum \left(R_t - \bar{R} \right)^2}{T - 1} \quad \sigma = \sqrt{\sigma^2}$$

Same idea as §2, but each past year gets equal weight. Average the returns, measure each year's gap from that average, square the gaps, add them up, and divide by **T − 1** (one less than the number of years).

Example. Stock M returned +12%, −4%, +18%, +7%, +2% over five years.

Average	(12 − 4 + 18 + 7 + 2) / 5 = **7%**
Gaps from 7%	+5, −11, +11, 0, −5
Squared gaps	25 + 121 + 121 + 0 + 25 = 292
Variance	292 / (5 − 1) = **73.00 %²**
Risk (σ)	√73 = **8.54%**

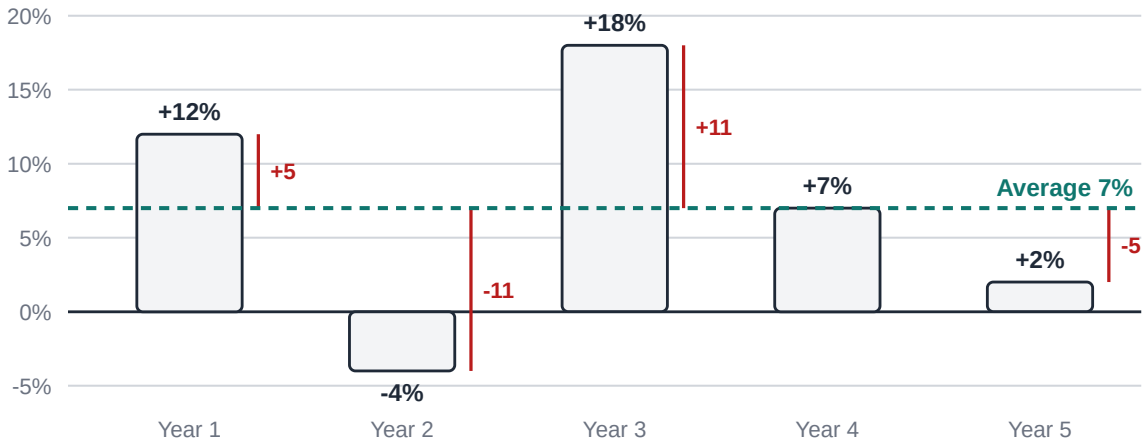


Figure 3. Stock M's five years. Red lines are each year's gap from the average; squaring and averaging those gaps gives the variance.

Watch out

- Divide by **T − 1** for history, not T. (§2 uses probabilities instead, so there's no division at all.)
- History is only a good guide to the future if the company and its world haven't changed much, and five years is a small sample.

4. Portfolio return: what a basket earned

I own several investments. What did the whole portfolio return?

$$w_i = \frac{\text{market value of holding i}}{\text{market value of the whole portfolio}}$$

$$R_p = w_1 R_1 + w_2 R_2 + \cdots + w_n R_n$$

A portfolio's return is the **weighted average** of its holdings' returns, where each weight is that holding's share of the portfolio's market value at the start.

Example. $10,000: 60 shares of Stock X at $100 ($6,000) and 200 shares of Stock Y at $20 ($4,000). A year later X is $106 and paid $1 per share (+7%); Y is $19 and paid nothing (−5%).

Weights	X: $6,000 / $10,000 = 60% · Y: $4,000 / $10,000 = 40%
Method 1: dollars	X profit 60 × ($6 + $1) = $420; Y profit 200 × (−$1) = −$200; total $220 / $10,000 = **2.2%**
Method 2: weights	0.60 × 7% + 0.40 × (−5%) = 4.2% − 2.0% = **2.2%**

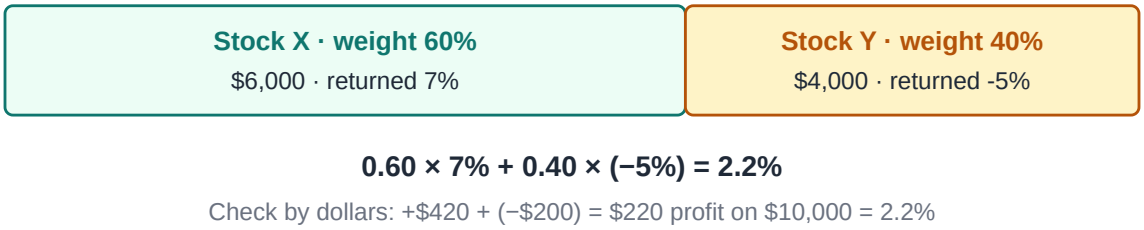


Figure 4. A portfolio's return is the weighted average of what each holding returned.

Watch out

- Weights come from **market values** (shares × price), not from the number of shares.
- Weights must add up to 100%.
- Weights drift on their own. By year-end X is worth $6,360 and Y $3,800, so the mix is about 62.6/37.4, not 60/40, even though you never traded.

5. Expected return of a portfolio

What should I expect my whole portfolio to earn next period?

$$E\left(R_p\right) = \sum w_i E\left(R_i\right)$$

Exactly the same weighted average as §4, using each holding's *expected* return instead of its actual one.

Example	70% in the stock from §2 (expected 9.50%) and 30% in a bond fund expected to return 4%: 0.70 × 9.5% + 0.30 × 4% = 6.65% + 1.20% = **7.85%**

> **Watch out**
>
> - This works for **return** only. Portfolio **risk** is *not* a simple weighted average; that's the whole point of §6.

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6. Correlation and portfolio risk

When I combine investments, how risky is the mix?

Portfolio risk depends on each holding's own risk *and* on whether they tend to move together. **Correlation** measures that, on a scale from −1 to +1: +1 means they always move in lockstep, 0 means no pattern, and −1 means they always move in opposite directions.

Step 1: covariance and correlation from past returns

$$\text{Cov}\left(A, B\right) = \frac{\sum \left(R_{A,t} - \bar{R}_A\right)\left(R_{B,t} - \bar{R}_B\right)}{T - 1}$$

$$r_{A,B} = \frac{\text{Cov}\left(A, B\right)}{\sigma_A \sigma_B}$$

Covariance multiplies the two investments' gaps from their averages, year by year. When both are above average (or both below) at the same time, the product is positive; when one is up while the other is down, it's negative. Dividing by both standard deviations rescales the result into a correlation between −1 and +1.

Example. Stock M (from §3: average 7%, risk 8.54%) and Stock N, which returned +4%, +8%, +6%, +3%, +4% (average 5%, risk 2.00%).

M's gaps	+5, -11, +11, +0, -5
N's gaps	-1, +3, +1, -2, -1
Products	(-5) + (-33) + (+11) + (-0) + (+5) = -22
Covariance	−22 / (5 − 1) = **-5.50 %²**
Correlation	−5.5 / (8.544 × 2) = **-0.322** (they lean toward moving opposite ways)

Step 2: portfolio risk

Two investments:

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B r_{A,B}\sigma_A\sigma_B$$

Three investments: one squared term per holding, plus one cross term for every pair.

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + w_C^2\sigma_C^2 + 2w_A w_B r_{\text{AB}}\sigma_A\sigma_B + 2w_A w_C r_{\text{AC}}\sigma_A\sigma_C + 2w_B w_C r_{\text{BC}}\sigma_B\sigma_C$$

Any number: keep going the same way, then take the square root.

$$\sigma_p = \sqrt{\sigma_p^2}$$

50/50 M + N	0.25 × 73 + 0.25 × 4 + 2 × 0.5 × 0.5 × (-0.322) × 8.544 × 2 = 18.25 + 1.00 − 2.75 = **16.50 %²**
Risk (σ)	√16.5 = **4.06%**
Check the long way	Build the 50/50 portfolio's return each year (+8%, +2%, +12%, +5%, +3%) and use §3 on it: variance 16.50 %², the same answer.
Three stocks	Equal thirds, each with 20% risk, correlations AB = +0.5, AC = −0.2, BC = +0.1: risk = **13.00%**

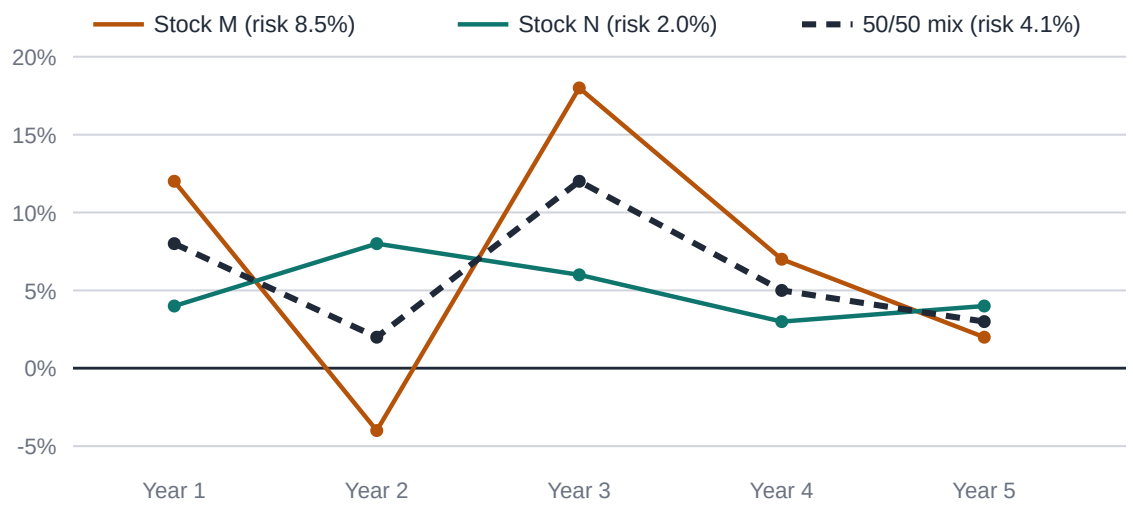


Figure 5. M and N tend to move in opposite directions (correlation -0.32), so the 50/50 mix is smoother than either average would suggest.

Diversification test. A portfolio is diversified when its risk is *lower* than the weighted average of its holdings' risks. Here: 4.06% vs. 0.5 × 8.54% + 0.5 × 2.00% = 5.27%. Diversified, and you gave up no expected return to get it.

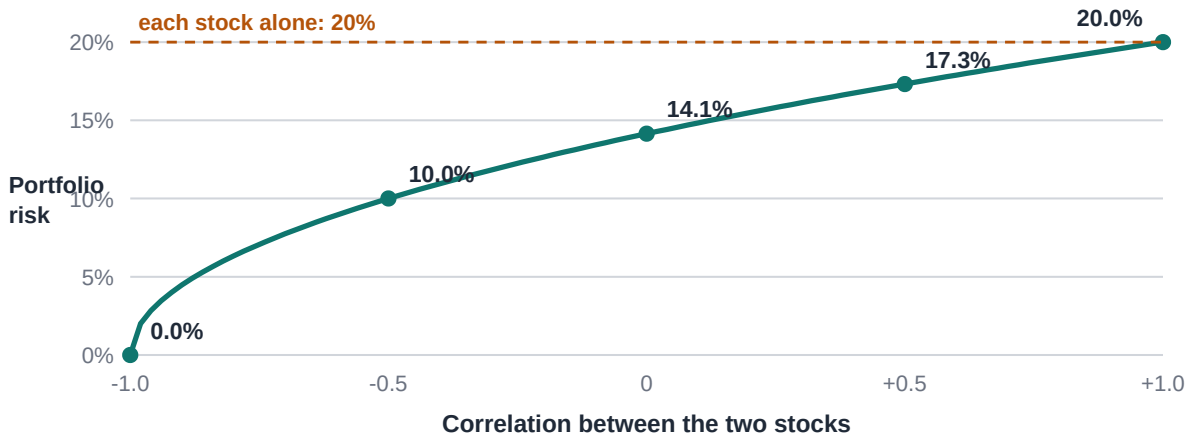


Figure 6. Correlation is the dial. Two stocks with 20% risk each, held 50/50: the lower their correlation, the lower the portfolio's risk. Expected return doesn't change.

Watch out

- Portfolio risk equals the weighted average of the risks **only** when correlation is exactly +1. Anything lower and mixing cuts risk.
- The cross term carries the correlation's sign. A negative correlation *subtracts* risk.
- Correlation must be between −1 and +1. If you get anything outside that range, there's an arithmetic error.

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7. Relevant risk: what a holding actually adds

Inside my portfolio, how much risk does each holding really contribute?

$$\text{Relevant risk of holding i} = \sigma_i \times \text{Corr}\left(R_i, R_p\right)$$

$$\sigma_p = \sum w_i \sigma_i \text{Corr}\left(R_i, R_p\right)$$

A holding's standard deviation is its **total** risk, measured on its own. Inside a portfolio, only the part that moves with the portfolio adds to its risk. Multiply the holding's risk by its correlation with the portfolio to get its **relevant risk**. Weight those and add them up, and you get the portfolio's risk exactly.

Stock M	risk 8.54% × correlation with portfolio 0.972 = relevant risk **8.31%**
Stock N	risk 2.00% × correlation with portfolio -0.092 = relevant risk **-0.18%**
Add them up	0.5 × 8.31% + 0.5 × (-0.18%) = **4.06%** = the portfolio's risk from §6

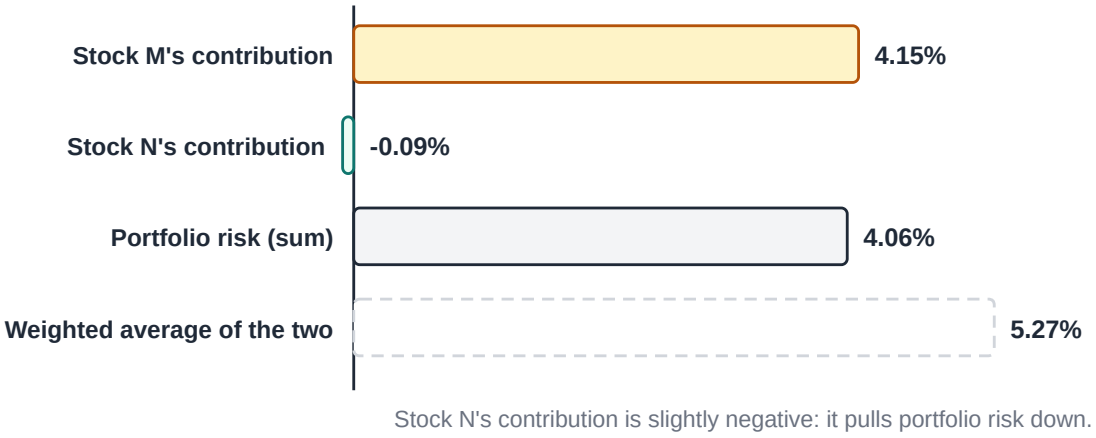


Figure 7. Relevant risk. Each holding's contribution is its weight × its risk × its correlation with the portfolio. The contributions add up to the portfolio's risk.

Watch out

- Relevant risk depends on the portfolio. Put the same stock in a different portfolio and its relevant risk changes.
- Relevant risk can be negative. That holding acts as a hedge: owning more of it (up to a point) lowers the portfolio's risk.
- A holding with high total risk can still add little to a portfolio if it doesn't move with the rest.

Quick reference

Question	Formula	§
What did it earn?	$(P_{end} - P_{start} + C) / P_{start}$	1
What should I expect (scenarios)?	$\Sigma\ p_s R_s$	2
How risky (scenarios)?	$\sqrt{\Sigma\ p_s(R_s - E(R))^2}$	2
Typical year (history)?	$(R_1 + \ldots + R_T) / T$	3
How risky (history)?	$\sqrt{[\Sigma\ (R_t - \bar{R})^2 / (T - 1)]}$	3
Portfolio return (actual or expected)?	$\Sigma\ w_i R_i$	4, 5
Do two holdings move together?	$Cov(A,B) / (\sigma_A \sigma_B)$	6
Two-holding portfolio risk?	$\sqrt{(w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B r_{AB}\sigma_A\sigma_B)}$	6
Is it diversified?	$\sigma_p < \Sigma\ w_i\sigma_i$	6
What does one holding add?	$\sigma_i \times Corr(R_i, R_p)$	7

Every number in this guide was generated and cross-checked by the companion script, *layerd_risk_return.py*.